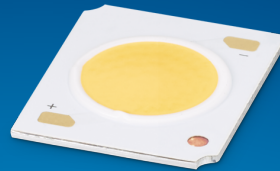


PHILIPS

Fortimo

LED

Fortimo SLM PW
1205 L13 2024 G7



Datasheet

Experience bright and vivid colors

Fortimo LED SLM PW 1205 L13 2024 G7

Fortimo LED SLM Gen7 continues to focus on the combination of Quality of Light and performance. By offering the CoB separate from the holder, even more flexibility in possible system combinations and specifications is achieved. This results in an extensive portfolio of lumen ranges, CCTs and spectra. Please also check the online Easy Design-in Tool for your perfect system combination (www.easydesignintool.com)

Key features and benefits

- Best quality of light available for all applications
- Extensive range of CCTs
- Small LES for narrow beam angles and small reflector designs
- Flexibility to select a different lumen output between 800 and 10000 lm
- State of the art Chip-on-Board (CoB) technology, enabling highest system efficacy
- System proposition (CoB + Holder + driver)
- Flexibility to optimize luminaire performance (lm/W or high lm output)
- Xitanium window drivers with SimpleSet for maximum flexibility
- Mini drivers for smallest possible luminaire designs
- Five years system warranty with over 50,000 hours lifetime
- Instant full light

March 2018

Ordering data

Commercial product name	EOC	12NC	Box quantity
Fortimo SLM C 830 PW 1205 L13 2024 G7	8718696 845875 00	9290 015 88606	20
Fortimo SLM C 930 PW 1205 L13 2024 G7	8718696 845905 00	9290 015 88706	20
Fortimo SLM C 935 PW 1205 L13 2024 G7	8718696 845936 00	9290 015 88806	20
Fortimo SLM C 940 PW 1205 L13 2024 G7	8718696 845967 00	9290 015 88906	20

Not all products are globally available by default.

Please contact your local Philips Lighting representative for local availability and activation.

Drive currents

Parameter	Nominal*	Life**	Max***	Unit
Fortimo SLM PW 1205 L13 2024 G7	500	870	1050	mA

Module temperatures

Parameter	Nominal*	Life**	Max***	Unit
T _c (case temperature at T _c point)	85	85	95	°C

* Nominal value at which typical performance is specified

** Value at which life time is specified

*** Maximum value for safe operation, do not operate above this value

Optical characteristics - table per color (CCT)

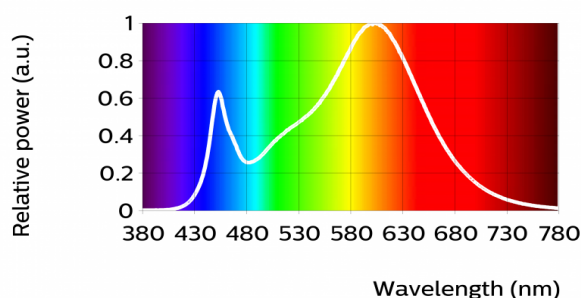
Fortimo SLM C 830 PW 1205 L13 2024 G7

Parameter	Min	Typ	Max	Unit
Luminous flux	2313	2570	2827	lm
Module efficacy	135	150		lm/W
Correlated color temperature (CCT)		3000		K
Color coordinates (CIEx, CIEy)		(0.422, 0.386)		-
Color consistency			3	SDCM
CRI	80	82		
Photobiological safety			RG1 unlimited	
Energy efficiency label		A++		

At currents higher than 991 mA the module might be classified as RG2

Measurement precision for flux +/- 5%. Measurement precision for efficacy +/- 6%. Measurement precision for x, y +/- 0.005. Measurement precision for CRI +/- 1.5

Operation point	830	lm	lm/W
80% I-nom 400mA	Tc 25 °C	2304	167
	Tc-nom 85 °C	2115	157
	Tc-max 95 °C	2082	155
I-nom 500mA	Tc 25 °C	2822	161
	Tc-nom 85 °C	2570	150
	Tc-max 95 °C	2526	148
I-life 1050mA	Tc 25 °C	5276	131
	Tc-nom 85 °C	4707	120
	Tc-max 95 °C	4608	117



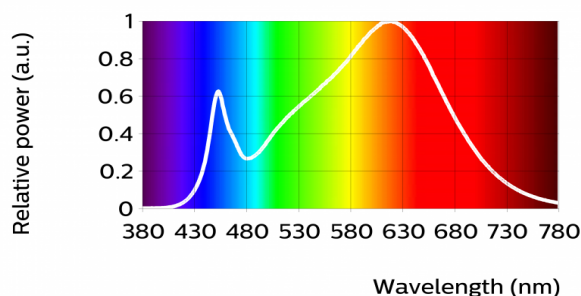
R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14
83	95	92	80	84	93	80	58	10	88	79	79	86	96

Fortimo SLM C 930 PW 1205 L13 2024 G7

Parameter	Min	Typ	Max	Unit
Luminous flux	1971	2190	2409	lm
Module efficacy	115	128		lm/W
Correlated color temperature (CCT)		3000		K
Color coordinates (CIEx, CIEy)		(0.422, 0.386)		-
Color consistency			3	SDCM
CRI	90	92		
Photobiological safety			RG1 unlimited	
Energy efficiency label		A++		

Measurement precision for flux +/- 5%. Measurement precision for efficacy +/- 6%. Measurement precision for x, y +/- 0.005. Measurement precision for CRI 1.5

Operation point	930	lm	lm/W
80% I-nom 400mA	Tc 25 °C	1963	143
	Tc-nom 85 °C	1803	134
	Tc-max 95 °C	1775	132
I-nom 500mA	Tc 25 °C	2404	137
	Tc-nom 85 °C	2190	128
	Tc-max 95 °C	2153	126
I-life 1050mA	Tc 25 °C	4493	112
	Tc-nom 85 °C	4008	102
	Tc-max 95 °C	3924	100



R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14
93	97	97	91	93	95	90	80	58	93	91	83	95	99

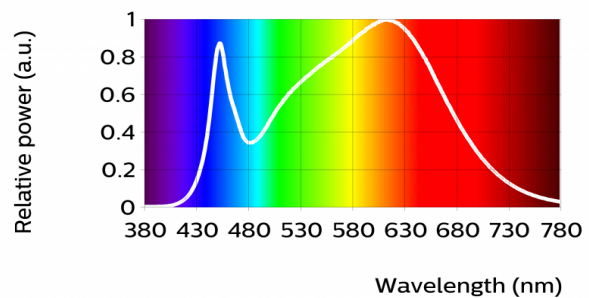
Fortimo SLM C 935 PW 1205 L13 2024 G7

Parameter	Min	Typ	Max	Unit
Luminous flux	2070	2300	2530	lm
Module efficacy	121	134		lm/W
Correlated color temperature (CCT)		3500		K
Color coordinates (CIEx, CIEy)		(0.398, 0.376)		-
Color consistency			3	SDCM
CRI	90	92		
Photobiological safety			RG1 unlimited	
Energy efficiency label		A++		

At currents higher than 916 mA the module might be classified as RG2

Measurement precision for flux +/- 5%. Measurement precision for efficacy +/- 6%. Measurement precision for x, y +/- 0.005. Measurement precision for CRI +/- 1.5

Operation point	935	lm	lm/W
80% I-nom 400mA	Tc 25 °C	2062	150
	Tc-nom 85 °C	1893	141
	Tc-max 95 °C	1864	139
I-nom 500mA	Tc 25 °C	2525	144
	Tc-nom 85 °C	2300	134
	Tc-max 95 °C	2261	132
I-life 1050mA	Tc 25 °C	4720	117
	Tc-nom 85 °C	4211	107
	Tc-max 95 °C	4122	105



R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14
93	96	97	91	92	93	92	82	59	89	91	77	94	98

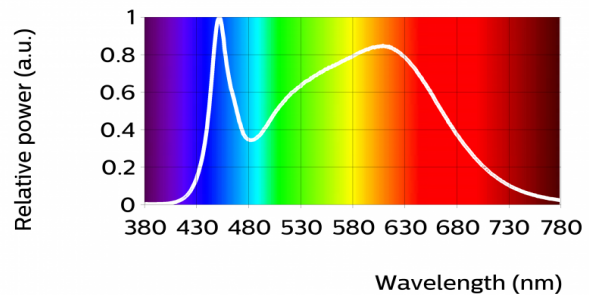
Fortimo SLM C 940 PW 1205 L13 2024 G7

Parameter	Min	Typ	Max	Unit
Luminous flux	2124	2360	2596	lm
Module efficacy	124	138		lm/W
Correlated color temperature (CCT)		4000		K
Color coordinates (CIEx, CIEy)		(0.374, 0.364)		-
Color consistency			3	SDCM
CRI	90	92		
Photobiological safety			RG1 unlimited	
Energy efficiency label		A++		

At currents higher than 745 mA the module might be classified as RG2

Measurement precision for flux +/- 5%. Measurement precision for efficacy +/- 6%. Measurement precision for x, y +/- 0.005. Measurement precision for CRI +/- 1.5

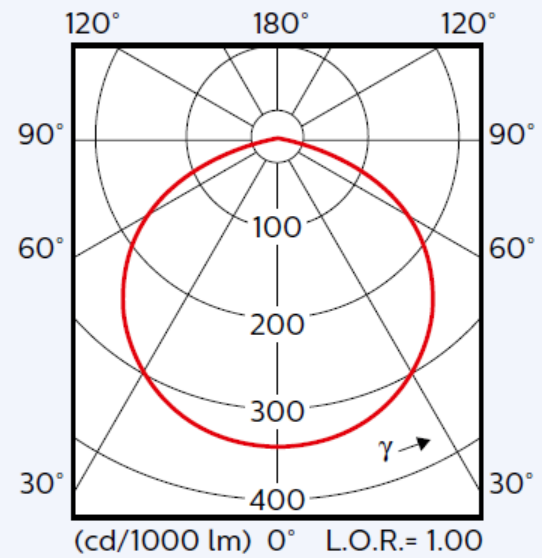
Operation point	940	lm	lm/W
80% I-nom 400mA	Tc 25 °C	2115	154
	Tc-nom 85 °C	1942	144
	Tc-max 95 °C	1912	143
I-nom 500mA	Tc 25 °C	2591	148
	Tc-nom 85 °C	2360	138
	Tc-max 95 °C	2320	136
I-life 1050mA	Tc 25 °C	4844	121
	Tc-nom 85 °C	4322	110
	Tc-max 95 °C	4231	108



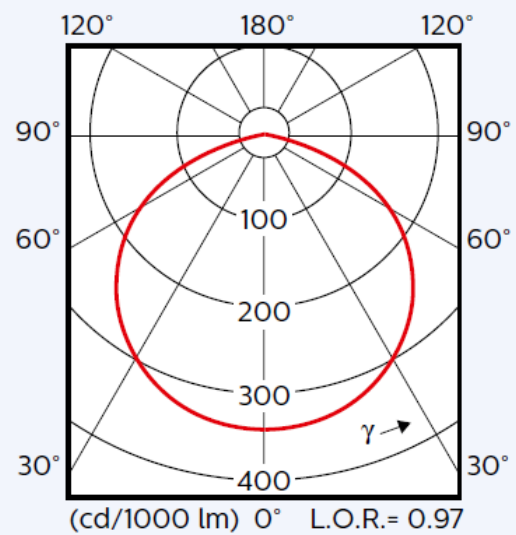
R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14
92	96	96	91	92	92	92	83	60	88	90	72	94	97

Beam shape

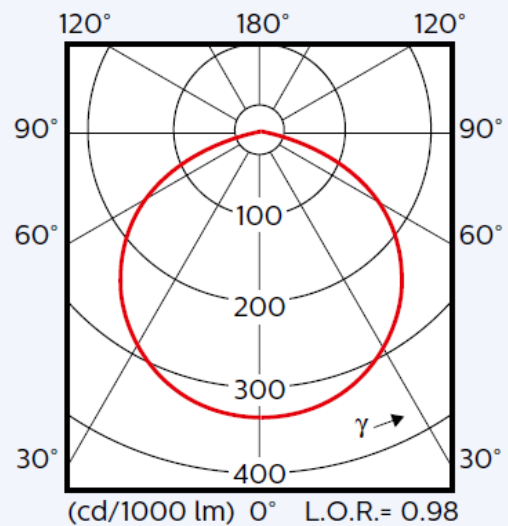
Bare CoB



CoB with a standard/ down-light/ Zhaga poke-in holder



CoB with a poke-in holder



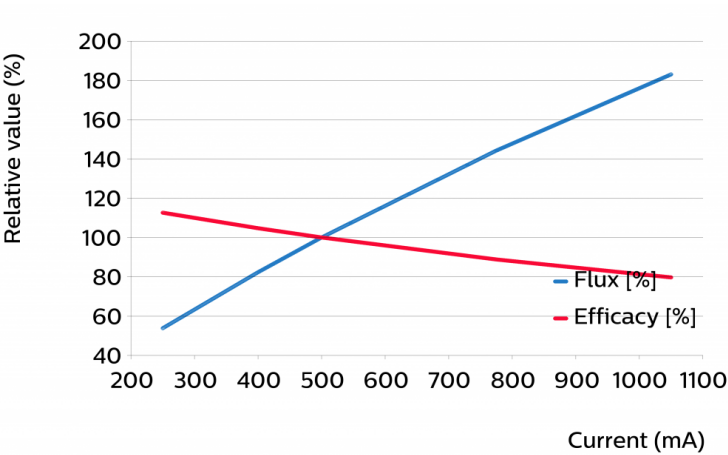
Fortimo SLM C 830 PW 1205 L13 2024 G7
Fortimo SLM C 930 PW 1205 L13 2024 G7
Fortimo SLM C 935 PW 1205 L13 2024 G7
Fortimo SLM C 940 PW 1205 L13 2024 G7

Parameter	Min	Typ	Max	Unit
Forward voltage	31.8	34.3	36.8	V
Power consumption	15.9	17.2	18.4	W

Tuning information

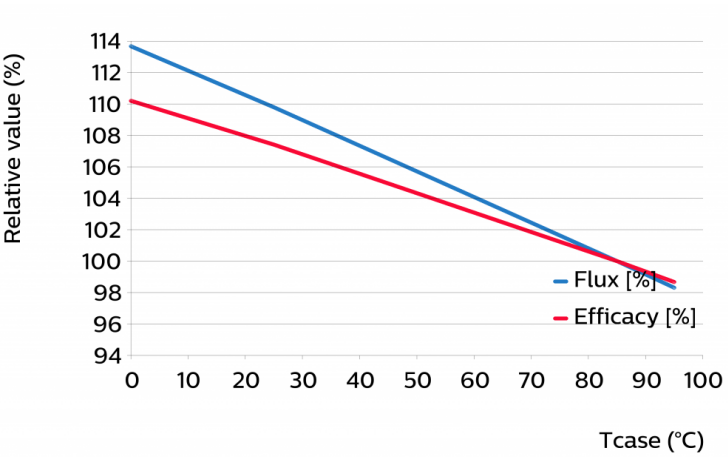
Flux and efficacy versus current (at Tc nominal)

I [mA]	Flux [%]	Efficacy [%]
1050	183	80
775	144	89
500	100	100
400	82	105
250	54	113



Flux and efficacy versus temperature at Tc (at I nominal)

Tcase [°C]	Flux [%]	Efficacy [%]
95	98	99
85	100	100
25	110	107
0	114	110



Lumen maintenance

Operation point	Lumen maintenance x 1000 hours	L70			L80			L90		
		B50	B20	B10	B50	B20	B10	B50	B20	B10
80% i nom 400 mA	Tc 65°C	>50	>50	>50	>50	>50	>50	>50	>50	41
	Tc life 85°C	>50	>50	>50	>50	48	39	35	23	18
	Tc max 95°C	>50	>50	42	>50	33	26	24	16	12
i nom 500 mA	Tc 65°C	>50	>50	>50	>50	>50	>50	>50	47	38
	Tc life 85°C	>50	>50	>50	>50	44	35	32	21	17
	Tc max 95°C	>50	48	39	46	30	24	22	14	11
I life 830 mA	Tc 65°C	>50	>50	>50	>50	>50	>50	49	32	26
	Tc life 85°C	>50	50	40	47	31	25	22	15	12
	Tc max 95°C	>50	35	28	33	22	17	16	10	8

Lifetime

Parameter	Value	Unit
C10 at Tc life	50000	hours

Please refer to the performance window to ensure that your operating conditions are covered.

>50 k hours claim is based on extrapolating raw LM80-data by using statistical techniques.

Switching cycles in accordance to EU 1194/2012: >15000.

Mechanical characteristics

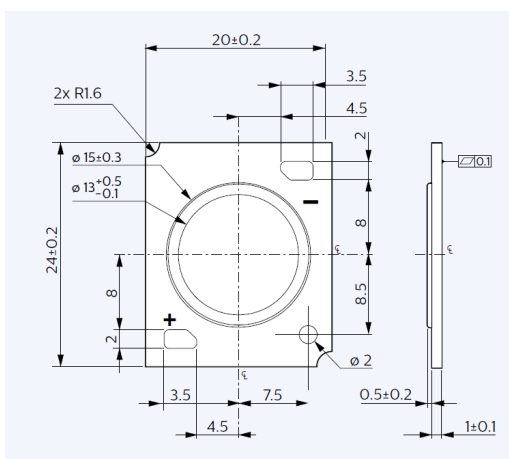
Fortimo SLM C 830 PW 1205 L13 2024 G7

Fortimo SLM C 930 PW 1205 L13 2024 G7

Fortimo SLM C 935 PW 1205 L13 2024 G7

Fortimo SLM C 940 PW 1205 L13 2024 G7

Parameter	Min	Typ	Max	Unit
Length	23.8	24	24.2	mm
Width	19.8	20	20.2	mm
Height PCB	0.9	1	1.1	mm
Height including dam	1.2	1.5	1.8	mm



Absolute ratings

Parameter	Min	Typ	Max	Unit
Current through the LED module (I-max)			1050	mA
Case temperature (Tc-max)			95	°C
Power at rated Vf-max and I-max			42	W
ESD (direct contact)			8	kV
Ambient temperature	-20		40	°C
Storage temperature	-40		80	°C

Application information

Certificates and Standards

IEC 62031:2008 (First Edition) + A1:2012 + A2:2014

EN 62031:2008 (First Edition) + A1:2013 + A2:2015

Relevant clauses of IEC 62471:2006 (Incl. IEC/TR 62471-2: 2009 and IEC/TR 62778: 2014)

Relevant clauses of IEC 60838-1:2004 + A1:2008 + A2:2011 with IEC 60838-2-2:2006 + A1:2012

Relevant clauses of EN 62471:2008 (With IEC/TR 62471-2: 2009 and IEC/TR 62778: 2014)

Relevant clauses of EN 60838-1:2004 + A1:2008 + A2:2011 with EN 60838-2-2:2006 + A1:2012

UL 8750

ENEC+

CE

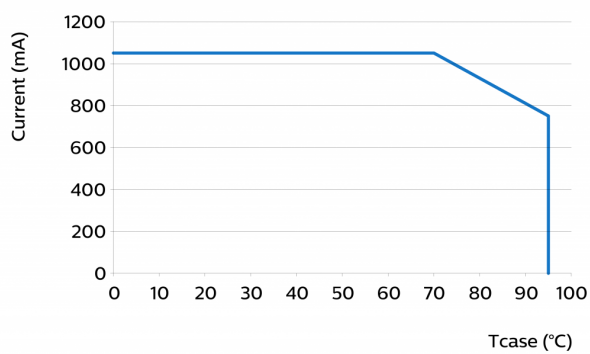
Environmental

RoHS/REACH

Application

IP rating	No IP-rating
Overheating protection	No
Luminaire class	IEC Class I and Class II
Dimming	Yes

Performance Window





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03/2018